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Notes on using Analog Devices DSP, audio, & video components from the Computer Products Division  
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## Introduction:

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The following Note was written by the Applications Engineers of White Mountain DSP, Inc.

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White Mountain DSP, Inc. is the leading third-party supplier of emulators and development systems for DSP technology today. White Mountain DSP's products provide easier and more cost-effective methods for engineers to develop and optimize DSP systems, shortening product development cycles for faster time-to-market.

White Mountain DSP covers the full range of SHARC® DSP processors and DSP-2106x solutions from Analog Devices. Emulators from White Mountain DSP are available for both PC and Sun workstation hosts providing development support for the Analog Devices ADSP-2106x SHARC® DSP processor family.

# Analog Devices JTAG Emulation Technical Reference

*Contributed by Larry D'Addario, White Mountain DSP, Inc.*

This application note describes how to design the interface between an Analog Devices SHARC DSP and the emulation header on a custom DSP target board. The environment described uses a White Mountain DSP emulator and an Analog Devices (ADI) SHARC DSP as an example. However, this information will apply equally well to any new JTAG DSPs from ADI.

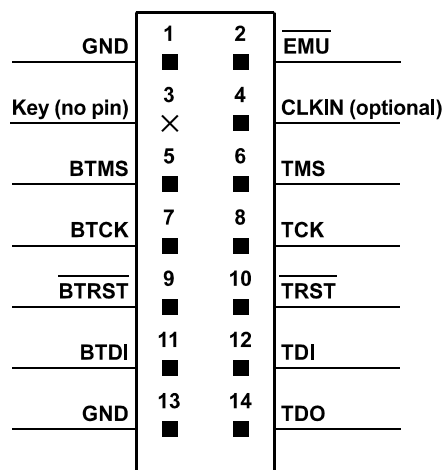
The White Mountain DSP family of emulators are tools that every SHARC DSP developer needs to test and debug their hardware and software system. ADI has supplied an IEEE 1149.1 JTAG Test Access Port (TAP) on each SHARC DSP. The emulator uses the JTAG TAP to access the internals of the DSP, allowing the developer to load code, set breakpoints, observe variables, observe memory, examine registers, etc.

The last thing an engineer wants to do when testing a new hardware design is debug the emulation port of their target board. Since the emulation port in many cases is the only vehicle to test a new target board, the emulation port needs to work flawlessly. The guidelines set forth in this application note are designed to help eliminate possible problems with the JTAG emulation port.

## TARGET BOARD CONNECTOR

The JTAG emulator interface to the SHARC DSP is a 14-pin header, as shown in figure 1. The customer must supply this header on their target board in order to communicate with the emulator. The interface consists of a standard dual row 0.025" square post header, set on 0.1" x 0.1" spacing, with a minimum post length of 0.235".

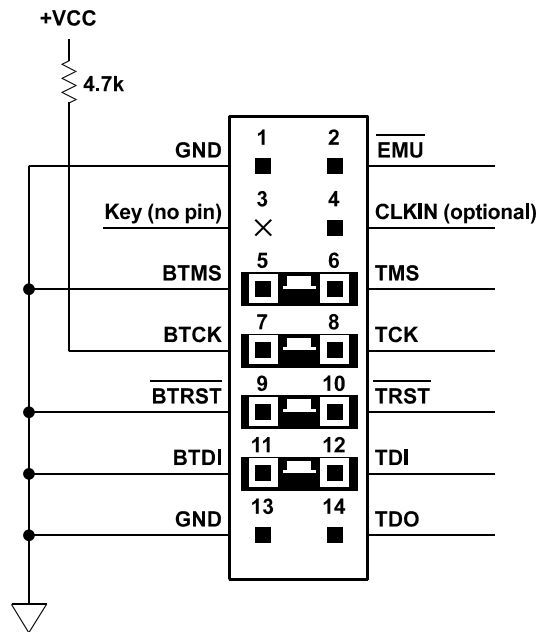
Pin 3 is the key used to prevent the pod from being inserted backwards. This pin must be clipped on the target board.



**Figure 1. JTAG Emulator Interface for SHARC DSP**

As can be seen in figure 1, there are two sets of signals on the header. There are the standard JTAG signals TMS, TCK, TDI, TDO, TRST, EMU, and CLKIN used for emulation purposes (via an emulator). There are also secondary JTAG signals BTMS, BTCK, BTRST, and BTDI that are optionally used for board-level (boundary scan) testing. The "B" signals would be connected to a separate on-board JTAG boundary scan controller if used. Most customers will never use the "B" signals. If they will not be used, tie all of them to ground except BTCK. BTCK should be pulled up to VCC (+5 or +3.3) using a 4.7KΩ resistor, as shown in figure 2.

When the emulator is not connected to this header, place jumpers across BTMS, BTCK, BTRST, and BTDI as shown in figure 2. This holds the JTAG signals in the correct state to allow the DSP to run free. Remove all the jumpers when connecting an emulator to the JTAG header.



**Figure 2. JTAG Target Board Connector With No Local Boundary Scan**

The state of each standard JTAG signal can be found in the following table.

**Table 1. State of Standard JTAG Signals**

| Signal           | Description         | Emulator | DSP              |
|------------------|---------------------|----------|------------------|
| TMS              | Test Mode Select    | O        | I                |
| TCK              | Test Clock (10 MHz) | O        | I                |
| TRST             | Test Reset          | O        | I                |
| TDI              | Test Data In        | O        | I                |
| TDO              | Test Data Out       | I        | O                |
| EMU              | Emulation Pin       | I        | I/O (Open Drain) |
| CLKIN (Optional) | DSP Clock Input     | I        | I                |

O = Output, I = Input, I/O = Input / Output

Note the CLKIN signal is presently not used by the emulator. If you are experiencing noise from this signal you can clip this pin on the 14-pin header. The CLKIN signal may be used for synchronous operations at a future date. For synchronous operation to work correctly, the CLKIN signal on all the DSPs and the emulator header must be the same signal and the skew between them must be minimal. If you have no plans for future synchronous operation, simply tie the CLKIN pin to ground on the 14-pin header (do not connect it to the DSP CLKIN signal).

The final connections between a single SHARC target and the emulation header (within 6 inches) are shown in figure 3. A 4.7KΩ pull-up resistor has been added on TCK, TDI, and TMS for increased noise resistance.

Should your design use more than one DSP (or other JTAG device), or if your JTAG header is more than 6 inches from the DSP, use a buffered connection scheme as shown in figure 4. Using a buffer that has built in series resistors such as the 74ABT2244 family will help quell ringing on the JTAG signal lines.

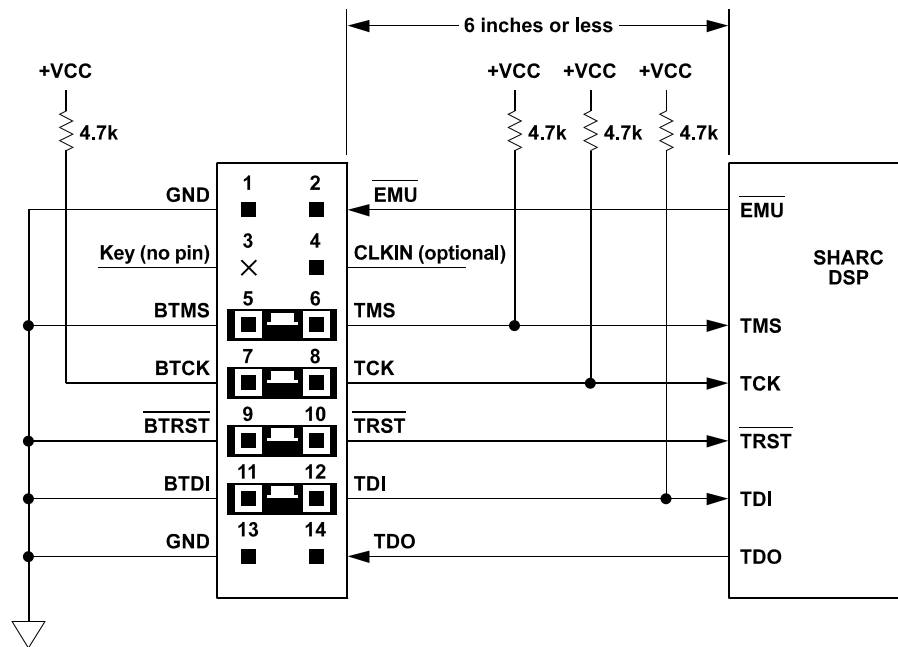


Figure 3. Single DSP Connection to JTAG Header

If you have more than one SHARC DSP on your target, it is imperative that you buffer the JTAG header. This will keep the signals clean and avoid problems that occur with longer signal traces.

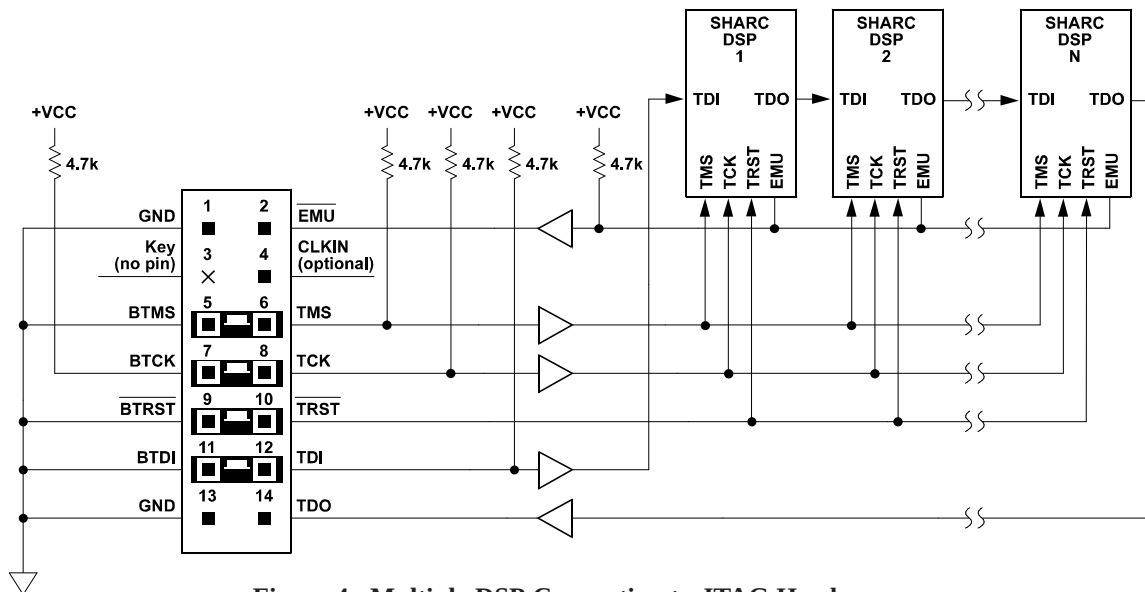


Figure 4. Multiple DSP Connection to JTAG Header

## LAYOUT REQUIREMENTS

All JTAG signals (TCK, TMS, TDI, TDO, EMU, TRST, CLKIN) should be treated as critical route signals. This means pay attention when routing these signals. Specify a controlled impedance requirement for each route (value depends on your circuit board - typically 50-75Ω). Keep crosstalk and inductance to a minimum on these lines by using a good ground plane and by routing away from other high noise signals such as clock lines. Keep these routes as short and clean as possible.

## JTAG POD LOGIC

A portion of the emulator pod interface is shown in figure 5. This figure describes the driver circuitry of the emulator pod. As can be seen, TMS, TCK, TRST, and TDI are driven with a  $10\Omega$  series resistor. TDO and CLKIN are terminated with an optional  $91/120\Omega$  parallel terminator. EMU is pulled up with a  $4.7K\Omega$  resistor. The 74LVT244 chip drives the signals at 3.3V, with a maximum current rating of  $\pm 32\text{mA}$ .

You can parallel terminate the TMS, TCK, TRST, and TDI lines locally on your target board, if needed, since they are driven by the pod with sufficient current drive ( $\pm 32\text{mA}$ ). In order to use the terminators on the TDO and CLKIN lines, you MUST have a buffer on your target board JTAG header and CLKIN signal. The SHARC DSP is not capable of driving the parallel terminator load directly with TDO. Assuming you have the proper buffers, you may use the optional parallel terminators simply by placing a jumper on J2 or J3.

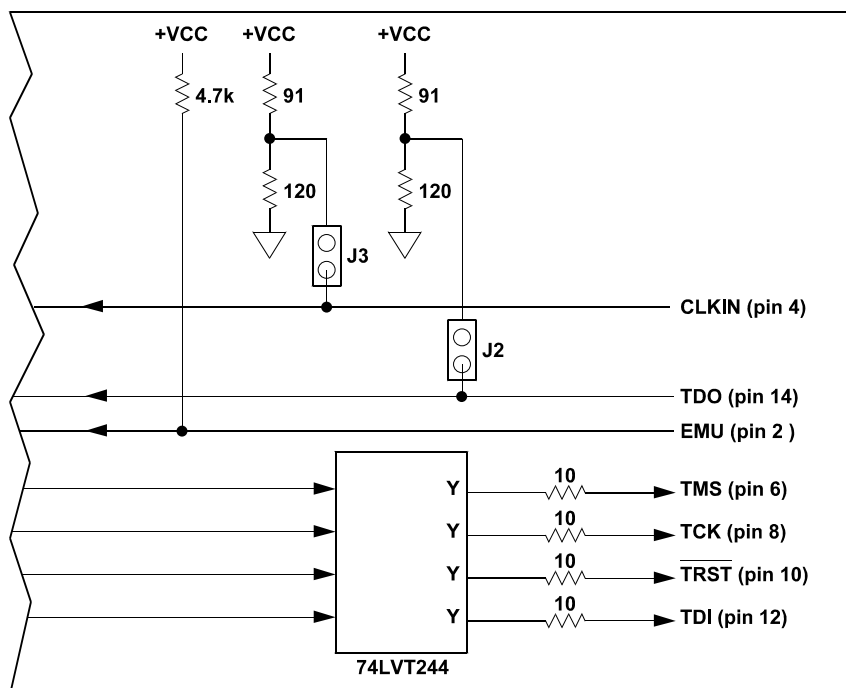


Figure 5. JTAG Pod Driver Logic

## POWER SEQUENCE

The power on sequence for your target and emulation system is as follows: apply power to the emulator first, then to the target board. Removal of power should be the reverse: turn off power to the target board then to the emulator. Upon power-on, the emulator drives the TRST signal low (releasing the DSP) until the emulation software takes control.

## CONCLUSION

Assuming the concepts described here are followed, designing the JTAG emulation interface for an Analog Devices SHARC DSP target board is an easy task, and the emulation interface should work with minimal effort. Valuable debug time can then be spent on the rest of the system (i.e. software and the rest of the hardware).



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